

Classifying Four Mystery Astronomical Objects Using Photometry, Parallax, Sky Position, and Optical Spectroscopy

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Submitted to ApJ

ABSTRACT

We classified four mystery astronomical objects using catalog photometry, Gaia astrometry, multi-wavelength imaging, sky-position analysis, and wavelength-calibrated spectra from the FAST spectrograph. The spectra identified emission and absorption features, which allowed us to compare them with standard stellar and galactic spectra. Object A’s high color index, infrared brightness, and strong TiO molecular absorption lines supported its classification as an M-type giant, most consistent with M5 III. Object B’s location on the plane of the Milky Way, spectral shape, and strong Balmer emission, especially in the H α band, supported its classification as a Classical T Tauri star and likely Class II young stellar object. Object C had a color index of 0, strong Balmer absorption, and early A-type spectral features, supporting its classification as an A0 III star. Object D lacked measurable parallax or proper motion, but displayed redshifted nebular emission lines upon further analysis, supporting its classification as a likely S0B spiral galaxy.

Keywords: stellar classification — spectroscopy — T Tauri stars — Stellar spectral types — parallax — young stellar objects — galaxies

1. INTRODUCTION

Astronomical objects can rarely be classified from a single measurement alone. Photometry can provide an initial estimate of an object’s color and apparent brightness, while parallax values can provide distance when the measurement uncertainty is small. Spectroscopy can be one of the most versatile classification tools because emission and absorption features reveal the physical conditions and composition of the source.

For stars, spectral classification is usually based on the Morgan-Keenan system, which connects spectral features to temperature and luminosity classes. Hot stars show helium and ionized metal lines, A-type stars show strong Balmer absorption, cooler F-K stars show

increasingly prominent metal lines, and M-type stars are marked by molecular bands such as TiO [Morgan \(n.d.\)](#), [Gray \(n.d.\)](#), [Zombeck \(1990\)](#). Most normal main-sequence stars show absorption-line spectra, while strong emission lines often indicate additional material or activity, such as accretion, winds, nebular gas, or star formation. These emission features are especially important for identifying Classical T Tauri stars, which commonly show strong H α from accreting circumstellar gas, and star-forming galaxies, which can show nebular lines such as H α , H β , O III, and S II ([Kurosawa et al. 2006](#); [Vogt n.d.](#)).

In this paper, we classify four mystery objects using a combination of Gaia astrometry and photometry, Johnson *B* and *V* magnitudes, multi-band imaging, RA/DEC sky-position analysis, and optical spectra from FAST. The FAST spectra are wavelength calibrated but not flux calibrated, which is a crucial limitation: their line positions can be used for classification, but their continuum shapes cannot be treated as physi-

cal spectral energy distributions. We therefore use parallax measurements where reliable to estimate distances and absolute magnitudes, and use color index values and spectral line identifications as our primary classification tools. This combined approach allowed us to classify Object A as an M5 III giant, Object B as a Classical T Tauri star and likely Class II YSO, Object C as an A0 III star, and Object D as a tentative S0B spiral galaxy.

2. DATA

The data set includes catalog, imaging, and spectroscopic measurements for Objects A, B, C, and D. The catalog measurements include approximate J2000 equatorial coordinates, Gaia parallaxes and proper motions where available, Gaia G , B_P , and R_P photometry, and Johnson B and V magnitudes where available. The Gaia astrometric measurements were used to estimate distances from parallax when the parallax uncertainty was sufficiently small, while the photometric measurements were used to calculate color indices and approximate absolute magnitudes (Wikipedia contributors n.d.a).

The imaging data span several wavelength regimes (and is located in Appendix A). Ultraviolet images were drawn from GALEX, covering approximately 135–280 nm where available (Wikipedia contributors n.d.b). Optical images were provided by DSS2, covering roughly 395–700 nm (Wikipedia contributors n.d.c). Near-infrared images came from 2MASS, covering approximately 1200–2100 nm (Wikipedia contributors n.d.d), and infrared images came from WISE/ALLWISE, covering approximately 3000–22,000 nm (Wikipedia contributors n.d.e). These multi-band images were used qualitatively to determine whether each object was most prominent in the ultraviolet, optical, near-infrared, or infrared.

The spectroscopic data were obtained with the FAST spectrograph and were provided as text files containing wavelength in Angstroms and flux counts (Harvard-Smithsonian Center for Astrophysics n.d.). These spectra are wavelength calibrated to a fraction of an Angstrom, allowing the positions of emission and absorption features to be compared with known atomic and molecular transitions. However, the spectra are not flux calibrated. As a result, the vertical scale and overall continuum shape are influenced by the detector response and observing conditions, and the spectra cannot be treated as proper spectral energy distributions. Therefore, we did not use the FAST data to compare continuum peaks or infer blackbody temperatures directly. Instead, the spectra are used primarily to identify the presence, absence, and approximate relative strengths of spectral lines.

Proper motion measurements were also available for some objects, but they were not used to derive distances. Proper motion alone measures angular motion across the sky, and converting that angular motion into a physical distance would require an independent estimate of tangential velocity. Since tangential velocities were not available for these objects, proper motion was used only as supporting contextual information rather than as a primary distance diagnostic. Essentially, if they were absent, it was an indicator that the object was so distant that its proper motion could not be detected.

2.1. Parallax Distances

For objects with reliable parallaxes, we estimated distance (d) through the relation,

$$d(\text{pc}) = \frac{1}{\text{parallax}(\text{arcsec})}. \quad (1)$$

Because the measured parallaxes are given in milliarcseconds, we first converted them using

$$p(\text{arcsec}) = \frac{p(\text{mas})}{1000}. \quad (2)$$

The asymmetric uncertainty range was calculated by inverting the upper and lower parallax values:

$$d_{\text{low}} = \frac{1}{(\text{parallax} + \sigma_p)/1000}, \quad (3)$$

$$d_{\text{high}} = \frac{1}{(\text{parallax} - \sigma_p)/1000}. \quad (4)$$

This method is appropriate for Objects B and C, whose fractional parallax uncertainties are relatively small. It is not reliable for Object A, because its parallax uncertainty was significantly larger than the parallax value itself.

2.2. Absolute Magnitudes

Where distance estimates were usable, absolute magnitudes were calculated using the distance modulus,

$$m - M = 5 \log_{10} \left(\frac{d}{10} \right). \quad (5)$$

Absolute magnitudes were then used as rough luminosity indicators. We also estimated luminosity in Johnson B and V bands in solar luminosities using

$$\frac{L}{L_{\odot}} = 10^{-0.4(M - M_{\odot})}. \quad (6)$$

These luminosity values are band-specific approximations rather than bolometric luminosities, as the data provided was insufficient to calculate a bolometric magnitude.

Table 1. General Object Measurements

Object	Parallax (mas)	RA (deg)	DEC (deg)	PMRA (mas/yr)	PMDEC (mas/yr)
A	0.019 ± 0.110	148.4	-1.8	-3.7 ± 0.2	-0.5 ± 0.2
B	2.493 ± 0.039	80.1	-5.8	1.15 ± 0.08	-1.29 ± 0.06
C	2.43 ± 0.11	77.5	10.0	1.20 ± 0.19	-0.33 ± 0.12
D	...	37.8	-9.8	...	...

Table 2. Photometric Measurements from GAIA and Johnson system B and V band magnitudes

Object	B Band	V Band	GAIA B	GAIA R	GAIA G
A	16.4	14.5	15.1	12.1	10.7
B	13.3	12.3	12.7	12.1	11.3
C	7.7	7.7	7.6	7.6	7.6
D	16.7	15.9	17.3	16.3	17.9

2.3. Color Index

The Johnson color index was calculated as

$$B - V = m_B - m_V. \quad (7)$$

A larger $B - V$ value corresponds to a redder object, while smaller or negative values correspond to hotter, bluer objects. We used $B - V$ as an initial guide to classify a potential spectral type letter, but not as the only classification tool, as reddening, circumstellar dust, and non-stellar emission can lead to distortions in the index's reliability.

2.4. Spectral Line Identification

The FAST spectra were plotted as flux counts versus wavelength. Because the spectra are not flux calibrated, we did not use their continuum shapes to infer temperatures or luminosities directly. Instead, we identified major spectral features by comparing the observed line positions with common stellar and nebular transitions. The most important features identified included hydrogen Balmer series lines, Ca II H and K, helium lines, metal lines, and molecular TiO bands. For Object D, redshifted emission features were compared with common galaxy emission lines such as H α , H β , O III, and S II.

3. ANALYSIS

3.1. Object A

RA and DEC values were converted into hours and arcminutes respectively in order to input them into the

sky model tool Aladin Lite (Bonnarel et al. 2000). This revealed a relatively cloud-free area of sky (at least optically), and the object itself sat outside of the plane of the Milky Way, meaning it was much less likely to be located near youthful planet-forming gaseous regions.

For Object A, the measured Gaia parallax is $p = 0.019 \pm 0.110$ mas. Although direct inversion of the central value gives a formal distance of $d \approx 5.26 \times 10^4$ pc, this value was not scientifically reliable because the parallax uncertainty was larger than the measured parallax itself. The fractional parallax uncertainty is $\sigma_p/p \approx 5.8$, and the one σ lower bound on the parallax is negative. Therefore, we do not adopt the inverted parallax distance as a secure distance estimate. Instead, we ignore this parallax value and instead use Object A's color and spectral features as the primary basis for classification. While this further limited the availability of data for this object, we felt that angling this object towards spectroscopic-heavy analysis and published classification spectral data would be more reliable for analytical credibility.

The photometry nonetheless was a viable tool to indicate color and a potential temperature spectral class for our object. The color index showed that Object A is very red, with a Johnson color index of

$$B - V = 16.4 - 14.5 = 1.90. \quad (8)$$

This value is consistent with an M-type stellar classification (Neuhäuser et al. 2022). The GAIA photometric bands and direct images (see Appendix A) also showed that the object is much brighter at redder wavelengths than at bluer wavelengths, further supporting this interpretation of a cool red object.

The spectrum of Object A shows broad absorption-like molecular features rather than narrow, isolated Balmer absorption lines. We classified clear emission and absorption lines for each object, as well as their corresponding elements, in Appendix B. Object A's spectra has a clear TiO molecular band progression, and appears to dramatically increase in the infrared. The large number of absorption lines also indicates a standard 'adult' main sequence star, or potentially a later

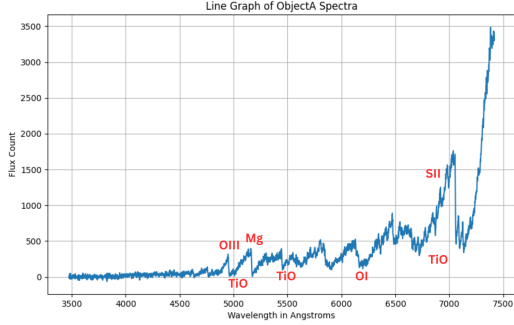


Figure 1. Object A Spectra, taken from FAST telescope data. Notable spectral features marked in red.

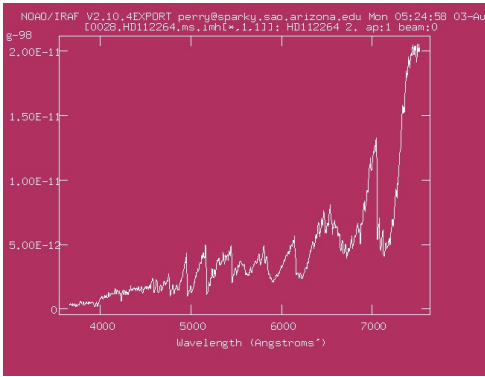


Figure 2. Comparable M5 III Spectral Graph. Image taken from (Berlind n.d.)

evolved star, but largely rules out a pre-main sequence object. TiO absorption is a defining characteristic of M-type stars (Gray n.d.; Zombeck 1990). The combination of a high $B - V$ value, strong red/infrared brightness, and TiO-dominated optical spectrum strongly supports an M-type classification.

If the formal distance were accepted, the inferred absolute magnitude would place Object A far above the main sequence, suggesting a giant or bright giant. However, because our Object A parallax measurement is highly uncertain and we decided to ignore it for this classification, the luminosity class cannot be established securely from distance. From here, we examined the spectra of several M-type stars to understand which spectral graphs were most similar in absorption lines, band width and general upticks or downturns as wavelength. Relevant spectra were derived from the Harvard atlas of established stellar spectra for varying spectral types Berlind (n.d.).

While bolometric magnitude analysis would have been preferred, there was no way with the given data to reasonably calculate this magnitude for a better ‘overall’ analysis across the varying bands.

While there is no precise scale to classify the subtypes based upon precise B-V values, we used a comparison to a common bright M-type object, Betelgeuse, which has a B-V of 1.85. Betelgeuse is classified as a M1-2 star (Keenan & McNeil 1989). Accordingly, with a slightly redder B-V value, we chose that a M2 classification would be a likely strong label, at least before spectral analysis.

Following this magnitude analysis and initial spectral type classification, the FAST spectral data was plotted in Python, as seen in Figure 1. All of the sub-zero values were not included in the graph, because they would be indicators of a calibration error on the part of the collecting instrument.

Notable wavelengths or wavelength bands and their corresponding emission Elements are displayed for Object A in Appendix B (analysis taken from Sloan Digital Sky Survey (n.d.a) and Sloan Digital Sky Survey (n.d.b)).

From these bands, we were able to identify a clear TiO series, further strengthening the hypothesis of a potential M-type star, as well as ionized oxygen, magnesium, and a Si II marker. These elements were marked in Figure 1. So many strong TiO markers places this as a strong M-type star, but more precise spectroscopic numbered subtype analysis comparing the relative strength of such lines was conducted through comparison of Harvard’s atlas of established stellar spectra for varying spectral types (Berlind n.d.).

While Object A’s color index more closely resembled that of Betelgeuse, a M1-2 I star, we believed that spectral shape and features would be more indicative of both temperature and potential stellar evolution, as luminosity could not be determined (Keenan & McNeil 1989). Object A displays clear emission dips, with broader emission ‘bands’ correlated with broader molecular sequences like the TiO series. In order to better clarify each band, estimations of each absorption peak or emission dip were estimated through Python’s graphing display feature that provides precise coordinates.

The M5 III star imaged resembles Object A’s spectra much better than a similar M1 or M2, largely due to the stark contrast in low to high wavelength flux count, as well as the clear progression of TiO lines. This graph also had a similar drastic increase in flux count past the 7500 Angstrom line, similarly to Object A.

3.2. Object B

Using the same Aladin Lite browser resource (Bonnarel et al. 2000), we used the given RA and DEC values to find Object B was located within or behind a dust cloud in a nebular region along the plane of the

288 Milky Way opposing the center of the galaxy. Under-
 289 standing that we are directly imaging through a cloud,
 290 there could be significant IR or emission excess to be
 291 mindful of.

292 For Object B, the measured parallax is $p = 2.493 \pm$
 293 0.039 mas. Converting from milliarcseconds to arcsec-
 294 onds and inverting the parallax gives

$$295 \quad d = \frac{1}{2.493/1000} = 401.1 \text{ pc.}$$

296 The lower parallax bound gives

$$297 \quad d_{\text{high}} = \frac{1}{2.454/1000} = 407.5 \text{ pc,}$$

298 while the upper parallax bound gives

$$299 \quad d_{\text{low}} = \frac{1}{2.532/1000} = 394.9 \text{ pc.}$$

300 Thus, we adopt a distance of

$$301 \quad d = 401.1^{+6.4}_{-6.2} \text{ pc,}$$

302 or approximately 401 ± 6 pc. Since the fractional par-
 303 allax uncertainty is only $\sigma_p/p \approx 1.6\%$, direct distance
 304 calculation is appropriate for this object.

305 With the general distance and location of the object
 306 within the sky configured, absolute magnitude for Ob-
 307 ject B was calculated using the following equation in
 308 Excel:

$$309 \quad m - M = 5 \log(d/10) \quad (9)$$

310 By inserting our estimated parallax value and appar-
 311 ent magnitude measurement, absolute magnitudes for
 312 each band could be better derived.

313 Absolute magnitude can also provide an indication of
 314 the theoretical luminosity of the object. Using a known
 315 absolute magnitude-luminosity relation, we could derive
 316 the relative luminosity of this object in solar luminos-
 317 ities. V and B-band absolute magnitude for the Sun was
 318 derived from (Kozłowski n.d.).

319 So, Object B is

$$320 \quad L/L_{\odot} = 10^{-0.4(M_B - M_B)} = 1.2 \quad (10)$$

321 times brighter than the sun's B-band magnitude! And
 322 is

$$323 \quad L/L_{\odot} = 10^{-0.4(M_v - M_v)} = 1.65 \quad (11)$$

324 times brighter than the Sun's V-band magnitude!
 325 Therefore, B is slightly more luminous than our Sun,
 326 but not significantly larger like a supergiant star would
 327 be.

Table 3. Object B Apparent and Absolute Magnitudes

Band	Apparent Magnitude	Absolute Magnitude
Johnson B	13.3	5.28
Johnson V	12.3	4.28
GAIA B	12.7	4.68
GAIA G	12.1	4.08
GAIA R	11.3	3.28

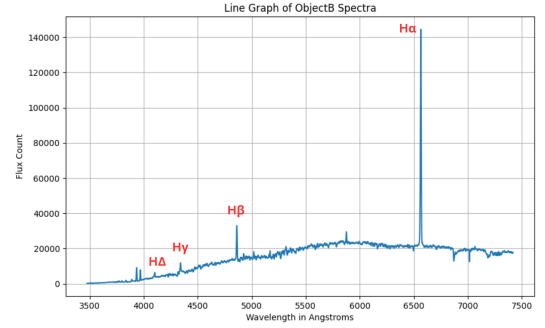


Figure 3. Spectra Data for Object B. Clear emission lines marked in red.

328 Using the same apparent to absolute magnitude equa-
 329 tion from Object A, and plugging in our respective ap-
 330 parent magnitudes and distance measurement, the pro-
 331 jected absolute magnitudes for each band were calcu-
 332 lated and are plotted in the table below. Note that
 333 these magnitudes are much more inherently reliable than
 334 those from Object A, and were used as a more significant
 335 identifier of stellar luminosity and spectral classification.
 336 Significant figures are the same across all objects.

337 Object B is relatively dim, at least in comparison to
 338 Object A. We identified a B-V color index of

$$339 \quad B - V = 13.3 - 12.3 = 1.00, \quad (12)$$

340 putting B in the region of a K-type orange star
 341 (Neuhäuser et al. 2022). Yet, with the spectra that we
 342 later examined, this object does not map with a typical
 343 K-type diagram.

344 After gaining this insight from our magnitude data,
 345 spectral data was graphed in Python.

346 We then noted the wavelength of every emission and
 347 absorption line that could be clearly identified, and
 348 placed those with a clear match (within 1-2 nm) in the
 349 table in Appendix B. Only the Balmer series were clear
 350 matches, with the H line being the clear strongest at 656
 351 nm.

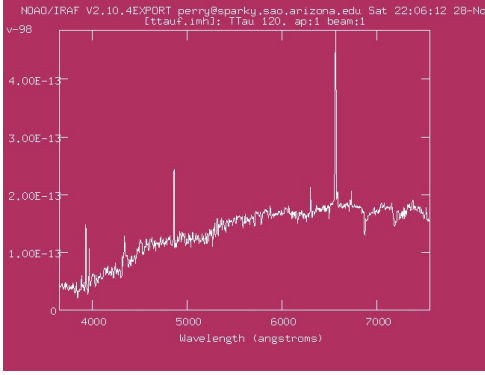


Figure 4. T Tauri Star Spectra

Table 4. Calculated Data for Object B

Data	Value
Distance	401.12 ± 6 pc
$B - V$	1.00
$L_V/L_\odot$	1.65
$L_B/L_\odot$	1.20

NOTE—Luminosity and distance measurements are rounded to three significant figures, and the $B - V$ value is rounded to three figures. The luminosities are band-specific estimates in solar luminosities.

So many emission lines and a lack of absorption lines indicate that this star is likely not yet fusing in its core, and is not a typical main sequence star, which would see strong absorption lines, particularly in K-type stars which this object would fall under from its color index alone. From this overall spectrum trend, young stellar object spectra seemed to be a better direction to search for comparable classification objects. In particular, we chose to examine the spectra of T Tauri stars, which are known for very strong H lines, which our object has. T Tauri are also known to be found along the star forming regions in the plane of the Milky Way, and to have a strong H line, and sometimes the remaining Balmer series (Nave n.d.; American Association of Variable Star Observers n.d.).

While this object is redder than our G-type Sun, it is slightly more luminous, placing somewhere above the main sequence on a typical Hertzsprung-Russell diagram. This further supports the idea that B may be a protostar headed down towards the main sequence, following the typical development cycle of a pre-main sequence star.

Having both the spectra, location, and luminosity pointing towards a pre-main sequence star, we examined several pre-main sequence stars' spectra until discovering a T Tauri graph shown in Figure 4 (Berlind n.d.). This graph matched our Balmer series line and relatively flat shape, overall increasing in flux count with wavelength, noting parallel spikes in the $H\alpha$, $H\beta$, emission around 390-397 nm, and absorption around 687 nm and 720 nm.

From this analysis, we cemented our initial classification of Object B as a T Tauri star with a K0 stellar temperature classification. Between Classical T Tauri Stars (CTTS) and weak-lined T Tauri stars, we labeled Object B a CTTS. According to (Kurosawa et al. 2006), a characteristic of a classical or strong-lined T-Tauri is the strength of the $H\alpha$ line, which this object possesses

strongly. This $H\alpha$ line, combined with the IR excess, makes it likely that there is disk accretion occurring, leading to a categorization of this object as a Classical T Tauri Star (CTTS). In terms of YSO classification, CTTS are correlated with Class II YSOs due to their disk accretion. This allowed us to ignore the traditional flux-based classification that most YSOs need to determine their specific object class.

3.3. Object C

For Object C, the measured parallax is $p = 2.43 \pm 0.11$ mas. Converting the parallax from milliarcseconds to arcseconds gives

$$d = \frac{1}{2.43/1000} = 411.5 \text{ pc.}$$

Because distance is inversely proportional to parallax, the upper and lower distance bounds are found by inverting the lower and upper parallax bounds, respectively. The lower parallax bound is $p - \sigma_p = 2.32$ mas, which gives

$$d_{\text{high}} = \frac{1}{2.32/1000} = 431.0 \text{ pc.}$$

The upper parallax bound is $p + \sigma_p = 2.54$ mas, which gives

$$d_{\text{low}} = \frac{1}{2.54/1000} = 393.7 \text{ pc.}$$

Therefore, we adopted a distance of

$$d = 411.5^{+19.5}_{-17.8} \text{ pc,}$$

or approximately 412 ± 19 pc. The fractional parallax uncertainty is $\sigma_p/p \approx 4.5\%$, which is larger than for Object B but still small enough that direct parallax inversion is a reasonable distance estimate.

Object C has nearly equal apparent magnitudes in all provided bands, with $B = 7.7$ and $V = 7.7$. Its color index is therefore

$$B - V = 7.7 - 7.7 = 0.00. \quad (13)$$

This color is consistent with a youthful high-mass star, near the boundary between late B and early A spectral types.

The absolute magnitude derived from the distance modulus is approximately

$$M_V \approx -0.37. \quad (14)$$

This is much brighter than a typical A0 V star, but the classification should be based primarily on the spectrum rather than on absolute magnitude alone.

The spectrum of Object C is dominated by absorption lines rather than emission lines, indicating a more normal stellar photosphere than in Object B. Strong Balmer absorption lines are present, including $H\delta$, $H\gamma$, $H\beta$, and $H\alpha$. The strength of the Balmer series is a classic indicator of A-type stars. Additional features, including He I and possible Ca II K, suggest that the object lies near the late-B/early-A boundary. Comparing these features with standard classification atlases favors an A0 classification (?).

The luminosity class is somewhat less certain. The strong absorption-line spectrum is compatible with a normal star, while the absolute magnitude suggests the object may be more luminous than a standard main-sequence A0 V star. We therefore classify Object C as an A0 star, likely A0 III or A0 V, with A0 III favored if the distance-based absolute magnitude is accepted.

Noting this similar distance to Object B, we used Aladin to find Object C sits on the middle plane of the Milky Way once more, but more on the furthest side. Once more, there could be a chance there are more youthful stars in this region due to the density of gas along the spiral arms of the galaxy, further strengthening our A0 hypothesis.

Given these absolute magnitudes, we once more plotted C's projected solar luminosity. Object C is

$$L/L_\odot = 10^{-0.4(M_V - M_V)} = 120 \quad (15)$$

times brighter than the Sun's V-band magnitude, and

$$L/L_\odot = 10^{-0.4(M_B - M_B)} = 219.17 \quad (16)$$

Table 5. Object C Apparent and Absolute Magnitudes

Band	Apparent Magnitude	Absolute Magnitude
Johnson B	7.7	-0.372
Johnson V	7.7	-0.372
GAIA B	7.6	-0.472
GAIA G	7.6	-0.472
GAIA R	7.6	-0.472

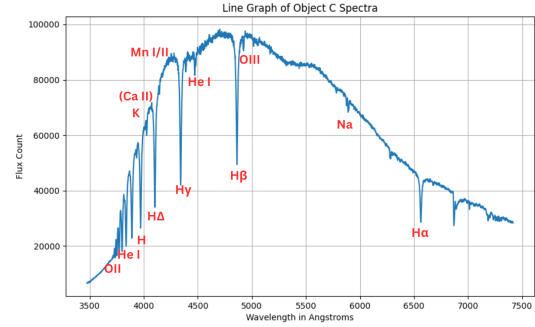


Figure 5. Object C Spectra, with absorption line elements marked in red.

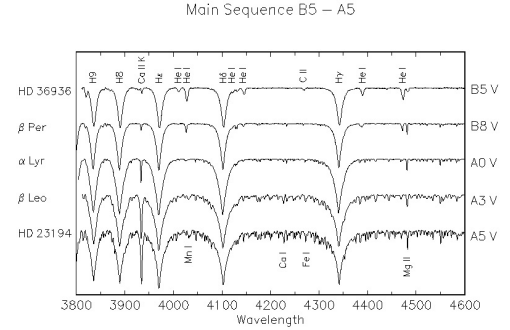


Figure 6. Image Used as Baseline Spectra for precise spectral type identification. Taken from Gray (n.d.).

times brighter than the Sun's B-band magnitude.

This luminosity would correspond with a main sequence B-type star, according to (Morgan n.d.) or potentially a highly luminous cooler star.

Examining the spectral graph provides further insights. There are many absorption lines in the graph which indicate a standard mains sequence star.

Due to the varying number of lines, and the identified late B-type or early A-type spectral classification, we analyzed varying spectra graphs within this range with a similar section of the wavelength band for Object C (Gray n.d.).

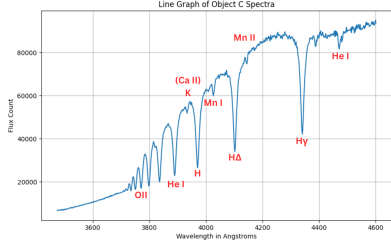


Figure 7. Python-generated snapshot of Wavelength band to compare with (Gray n.d.) absorption lines.

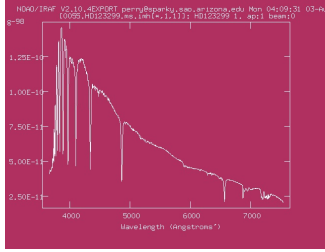


Figure 8. Best Object C comparison A0 III star from (Berlind n.d.)

Because the comparison spectra and the observed FAST spectrum are not displayed on the same flux scale, the classification was based primarily on the relative depths and presence of particular absorption lines rather than on the overall continuum shape. The Ca II K feature appears relatively weak, which is most consistent with a late-B to early-A classification. The Mn I feature is more consistent with an A0-A3 spectrum, while the He I line strength suggests a classification near B8-A0. The overall density and depth of the absorption features also most closely resemble spectra in the A0-A3 range. Because the Balmer absorption features are strong and the majority of the identified line strengths are most consistent with the early-A sequence, we decided to adopt an A0 classification for Object C. The dominance of absorption features, rather than emission features, supports the interpretation that Object C is a normal stellar photosphere and is therefore likely a mature A0 star on the main sequence or having left it.

After determining an A0 classification, we once again used the Harvard atlas (Berlind n.d.) for specific spectral type classification based on lines present. While A0 stars III, IV, and V all displayed promising similarities, we eventually settled on A0 III for its lack of Na line and weaker absorption dips between the Balmer series lines compared to the main sequence star. However, it must be stressed that all three graphs shared many similarities and, having been given more precise flux data, a more precise luminosity or temperature approximation

Table 6. Calculated Data for Object C

Data	Value
Distance	411.52 ± 18 pc
$B - V$	0.0
$L_V/L_\odot$	120
$L_B/L_\odot$	219

NOTE—Distance and luminosity were rounded to approximately three significant figures, and the $B - V$ index is rounded to one decimal place.

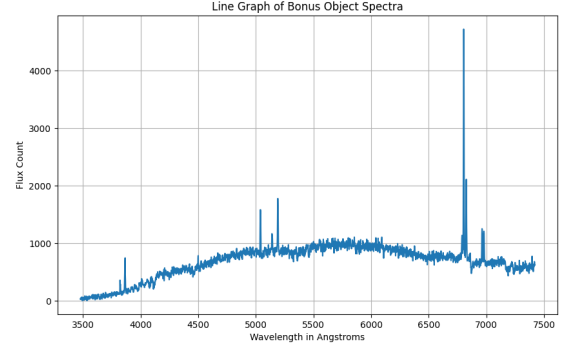


Figure 9. Object D Spectra plotted from FAST data.

would have aided this decision and potentially lead to a different classification.

3.4. Object D

This object is distinct in that it does not contain values for parallax and proper motion measurements, and, upon RA and DEC location analysis in Aladin, was found to exist nearby several other faint galaxies. Because parallax and proper motion measurements were not available for this object, it is likely that it is so far away in distance that it could not be detected via GAIA's precision for parallax measurement.

This led us to go directly to spectral analysis, using the same Python plot from the FAST spectral data.

Upon initial plotting of spectral lines, no matches were found. This was due to a detected redshift of lines, causing emission lines to plot 25 nanometers higher than they would otherwise be located. The presence of such a red shift leads one to further assume a large structure such as a galaxy that is extremely distant was being plotted, or potentially a very bright but distant star. Several

strong observed features can be matched to common nebular emission lines if this redshift correction is applied. The most important lines include $H\beta$, O III, $H\alpha$, and S II (See Appendix B for all lines identified). For example, $H\alpha$ has a rest wavelength of approximately 656 nm and appears near 681 nm in the observed spectrum.

The spectrum is dominated by emission lines rather than by a clean stellar absorption sequence. This suggests ionized gas associated with star formation. Star-forming galaxies commonly show strong Balmer series, oxygen and sulfur emission lines (Vogt n.d.). The presence of these lines, combined with the redshifted light, supports a galaxy classification rather than a stellar classification.

We ruled out a potential protostar or gas-forming region due to the absorption line ‘noise’ present throughout the data. In a spectra of our Object B T Tauri star, the same emission lines are present, but there are notably a lack of absorption lines, since nuclear fusion has not yet begun in the protostar properly. Examining a sample spiral galaxy spectra confirmed the likely reason for this ‘noise’: the gaseous young stars in the spiral arms heat their surrounding gas, and the conglomeration of underlying older stars create small absorption features throughout the rest of the spectra Vogt (n.d.). The spectrum is more consistent with a spiral galaxy than with an elliptical galaxy, because the latter of which would generally show a smoother, older stellar absorption spectrum with weaker nebular emission.

Having classified this object as a spiral galaxy, we used the de Vaucouleurs extension of the Hubble sequence classification model for galaxies to better categorize the object. Admittedly, the morphology is difficult to determine precisely from the available images. However, examining the object in the UV imaging band (pictured in Appendix A), one can see a generally oblong circle. We therefore classified Object D as a tentative S0B-type spiral galaxy.

4. RESULTS

The final classifications are summarized in Table 7. Object A is likely a cool M-type star with strong TiO features. Object B is a classical T Tauri star and likely Class II YSO. Object C is an early A-type star with strong Balmer absorption. Object D is best interpreted as a redshifted S0B spiral galaxy.

5. DISCUSSION

The classifications in this study demonstrate the importance of combining several different diagnostics to

reach desired outcomes in spectral type classification. The most important methodological limitation was that our FAST spectra was not flux calibrated. As a result, the apparent continuum height, slope, and peak position were unable to be interpreted as a standard spectral energy distribution. The spectra remained useful, however, because they were wavelength calibrated, allowing line positions and approximate relative line strengths to be compared with known stellar and nebular transitions.

Object A illustrated the danger of relying too heavily on distance-based quantities when the parallax uncertainty is large. Its Gaia parallax, $p = 0.019 \pm 0.110$ mas, had an uncertainty far greater than the measured value itself. Consequently, absolute magnitudes and luminosities derived from this distance could not be used as reliable classification evidence, and thus parallax values were discarded for this object. Instead, the classification of Object A rested primarily on photometric color and spectral morphology. Its large $B - V$ value of 1.90 showed that it was a very red object, and its stronger appearance at redder and infrared wavelengths supported the interpretation that it was a cooler star. Spectroscopically, Object A showed broad molecular absorption features, especially a clear TiO progression, which is characteristic of M-type stars. Comparison with reference spectra suggests that the observed spectrum more closely resembles an M5 III giant than the initially considered earlier M-type classifications. Therefore, Object A was best classified as an M5 III giant, although its luminosity class remains less secure than it would have been with a reliable parallax or flux-calibrated spectrum.

Object B presented a different case because its parallax is reliable. Its distance of $401.1^{+6.4}_{-6.2}$ pc is well constrained, with a fractional parallax uncertainty of only about 1.6%. Its $B - V = 1.00$ color index initially suggests a K-type star, and its absolute magnitudes indicate that it is only moderately luminous compared with the Sun. However, the spectrum does not resemble a normal K-type main-sequence photosphere. Instead of being dominated by absorption lines, Object B shows strong Balmer emission, especially $H\alpha$ near 656 nm. A standard K-type main-sequence star would be expected to show many neutral metal absorption features, not a spectrum dominated by hydrogen emission. The emission-line spectrum, apparent infrared excess, and location in or behind a dusty region of the Milky Way point toward a young stellar object. Because strong $H\alpha$ emission is a defining feature of accreting Classical T Tauri stars, Object B is best interpreted as a Classical T Tauri star. Its likely disk-related infrared behavior further supports identifying it as a Class II YSO.

Table 7. Summary of Derived Properties and Final Classifications

Object	Distance	$B - V$	Key Spectral Evidence	Proposed Classification
A	Technically around 52.6 kpc, unreliable	1.90	TiO molecular bands; very red color	Tentative M5 III giant
B	$401.1^{+6.4}_{-6.2}$ pc	1.00	Strong Balmer emission, especially H α	Classical T Tauri star; likely Class II YSO
C	$411.5^{+19.5}_{-17.8}$ pc	0.00	Strong Balmer absorption; He I; weak Ca II K	A0 III/V star
D	...	...	Redshifted H α , H β , O III, S II emission	Star-forming spiral/disk galaxy S0B

Object C is the most straightforward stellar photosphere in the sample. Its parallax provided a distance of $411.5^{+19.5}_{-17.8}$ pc, which was slightly less precise than Object B but still reliable enough for direct inversion. Its $B - V = 0.00$ color index placed it near the late-B/early-A boundary. The spectrum was dominated by absorption lines rather than emission lines, supporting the interpretation of a mature stellar photosphere. Strong Balmer absorption, along with He I and relatively weak Ca II K, pointed toward an early-A classification. Comparison with reference spectra in the B8-A3 range favored A0 overall. The luminosity class was more difficult to determine: its absolute magnitude is brighter than a typical A0 V star, and comparison with A0 III, IV, and V spectra showed similarities across multiple luminosity classes. The weaker absorption structure between Balmer lines and the lack of a strong Na feature made A0 III the best fit among the available comparisons. Thus, Object C was classified as A0 III, though A0 IV or A0 V are difficult to fully exclude without higher-quality flux-sensitive spectral data.

Object D differed from the other objects because it lacked Gaia parallax and proper motion measurements and appeared near other faint galaxies in the sky. This already suggested an object too distant for Gaia parallax detection rather than a nearby star. Our hypothesis was further strengthened when examining D's spectrum. Initially, the strongest emission features did not align with commonly identified emission wavelengths; after recognizing an approximate redshift of about 25 nm, however, the lines matched common nebular transitions including H α , O III, and S II. Such redshifted emission would not be expected from a nearby normal star and instead indicated an extragalactic source. The combination of strong emission lines and smaller absorption features was consistent with a galaxy containing both ionized gas from star-forming regions and older underlying stellar populations. This supported classification as a star-forming spiral rather than an elliptical galaxy, which would generally show weaker nebular emission and a smoother old-stellar absorption spectrum. The morphology was difficult to determine from the avail-

able images, but the oblong appearance in the imaging data led to a tentative S0B spiral classification. Because the imaging resolution was limited, the exact identified structure should be treated as uncertain.

The primary sources of uncertainty were lack of flux-calibrated data, parallax error measurements for A and C, line blending or potential misidentification, atmospheric or instrumental error effects, and the limited resolution of imaging data. Future work with flux-calibrated spectra, larger reference-spectrum libraries, and higher-resolution imaging would likely greatly improve the classification interpretation of all four objects.

6. CONCLUSION

We classified four mystery astronomical objects by combining photometry, Gaia astrometry, sky-position analysis, multi-wavelength imaging, and wavelength-calibrated optical spectroscopy. Object A is best classified as an M5 III giant: although its parallax was too uncertain to support a reliable distance or luminosity estimate, its red $B - V = 1.90$ color, infrared brightness, and TiO-dominated spectrum all support a cool M-type giant classification. Object B is best classified as a Classical T Tauri star and likely Class II young stellar object. Its reliable distance of approximately 401 pc, red color, dusty Milky Way environment, likely infrared excess, and strong Balmer emission, especially H α , point toward an accreting pre-main-sequence object rather than a normal K-type main-sequence star.

Object C is best classified as an A0 III star, based on its distance of approximately 412 pc, $B - V = 0.00$ color index, strong Balmer absorption, and similarity to early-A reference spectra. Object D is best classified as a tentative S0B spiral galaxy because it lacks measurable parallax or proper motion, appears near other faint galaxies, and shows redshifted nebular emission lines consistent with an extragalactic star-forming source. The largest limitation of this analysis is that the FAST spectra were not flux calibrated, so they could not be treated as true SEDs or used to derive reliable temperatures, bolometric luminosities, or radii. Instead, the strongest classifications came from spectral line positions, line strengths,

color indices, and agreement across multiple independent diagnostics.

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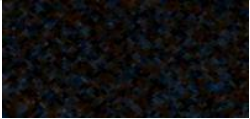
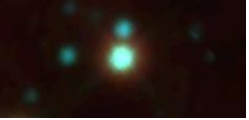
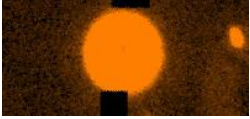
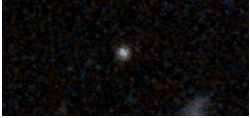
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APPENDIX

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A. MULTI-WAVELENGTH IMAGING DATA

Table 8. Multi-wavelength imaging data for the mystery objects. Images are shown in UV, optical, near-infrared, and infrared bands where available.

Object	UV: GALEX 135–280 nm	Optical: DSS2 395–700 nm	Near-IR: 2MASS 1200–2100 nm	IR: ALLWISE 3000–22000 nm
A				
B	Not available			
C				
D				

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B. MAJOR SPECTRAL FEATURES IDENTIFIED FROM OBJECT WAVELENGTH

Table 9. Object A Spectral Features

Element	Corresponding Wavelength (nm)	Object A Wavelength (nm)
TiO / OIII	490-520 / 493	496
TiO / Mg	490-520 / 518	518
TiO	540-570	545
O I	637	637
TiO / SiI	670-690 / 672	672
TiO	670-690	679
TiO	670-690	687
TiO	670-690	690

Table 10. Object B Spectral Features

Element	Corresponding Wavelength (nm)	Object B Wavelength (nm)
H	410	410
H	434	434
H	486	486
H	656	657

Table 11. Object C Spectral Features

Element	Corresponding Wavelength (nm)	Object C Wavelength (nm)
O II	373	374
He I	389	389 (huge)
H	397	397 (huge)
H	410	410 (huge)
H	434	434 (huge)
K (maybe Ca II K)	393	394
Mn I	403	403
Mn II	414	414
He I	447	447
H	486	486 (huge)
O III	493	492
Na	589.56	589
H	656	656 (large)

Table 12. Redshift-Corrected Spectral Features for Object D

Element	Corresponding Wavelength (nm)	Object D Wavelength with Redshift Subtracted Out (nm)
O III	496	495
H	656	656
S II	672	672
S II	673	673